Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2025

Course: B.Tech
Code: 100607

Subject: Introduction to VLSI Design

Time: 03 Hours
Full Marks: 70

Instructions:-

(i) The marks are indicated in the right-hand margin.
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
(iv) Question No. 1 is compulsory.

Q.1 ***Choose the correct option of the following (Any seven question only):*** [2 x 7 = 14]

(a) MOSFET capacitances depend on
- (i) Channel doping only
- (ii) Biasing conditions
- (iii) Layout rules
- (iv) Mobility of carriers

(b) Bipolar transistors offer
- (i) High input impedance
- (ii) Low noise margin
- (iii) High speed and drive strength
- (iv) High leakage

(c) BiCMOS consumes more power than CMOS because
- (i) Bipolar devices have static bias currents
- (ii) High interconnect parasitics
- (iii) CMOS scaling issues
- (iv) Low threshold voltages

(d) SRAM cell is typically built using
- (i) 2 transistors
- (ii) 6 transistors
- (iii) 4 transistors
- (iv) 8 transistors

(e) Arithmetic circuits include
- (i) Shifters
- (ii) Registers
- (iii) Adders & multipliers
- (iv) Both (i) & (iii)

(f) PTL stands for
- (i) Pseudo Transistor Logic
- (ii) Power Tuned Logic
- (iii) Pass Transistor Logic
- (iv) Pre-charge Ternary Logic

(g) TSPC registers are used for
- (i) Multi-phase clocking
- (ii) Leakage reduction
- (iii) Single-phase high-speed clocking
- (iv) Memory design

(h) In Complementary CMOS logic, power is mainly consumed during
- (i) Static state
- (ii) Body biasing
- (iii) Switching activity
- (iv) Leakage only

(i) Low-power design techniques target reduction of
- (i) Channel length
- (ii) Parasitic capacitances only
- (iii) Switching activity and supply voltage
- (iv) Gate count

(j) Supply voltage scaling mainly affects
- (i) VTC shape
- (ii) Layout parasitics
- (iii) Speed and noise margin
- (iv) Channel doping

Q.2 Explain the MOSFET threshold voltage and derive expressions for the factors affecting it including body bias effect. [14]

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Q.3	(a)	Derive the static characteristics and noise margins of a CMOS inverter.	[7]
	(b)	Explain latch-up in CMOS, methods of prevention and layout considerations.	[7]
Q.4	(a)	Discuss propagation delay estimation using the Elmore delay model.	[7]
	(b)	Describe low-power design techniques in CMOS circuits.	[7]
Q.5	(a)	Explain pseudo-NMOS logic and compare it with CMOS logic.	[7]
	(b)	Discuss Domino logic operation and its derivatives.	[7]
Q.6	(a)	Describe the design and working of a CMOS shifter.	[7]
	(b)	Compare SRAM and DRAM in terms of speed, area, leakage and applications.	[7]
Q.7	(a)	Explain the design challenges and applications of BiCMOS logic circuits.	[7]
	(b)	Discuss static and dynamic behaviour of BiCMOS circuits.	[7]
Q.8	(a)	Explain CMOS layout elements, design rules and parasitics due to wires and vias.	[7]
	(b)	Discuss channel-length modulation and short channel effects in deep sub-micron MOSFETs.	[7]
Q.9		Explain the effect of process variation and supply scaling on inverter performance.	[14]

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